
Chapter 29 – D.C. Motors (Full Detailed Notes)

1. Introduction

- A **DC motor** is a machine that converts **electrical energy (direct current)** into **mechanical energy (rotation of shaft)**.
- It is essentially the same machine as a **DC generator**, but operated in reverse.
- In fact, any DC machine can act either as a **generator** or as a **motor**, depending on whether it is **driven mechanically or electrically**.

2. Principle of Operation

- A conductor carrying current, when placed in a magnetic field, experiences a mechanical force.
- This is the **motor principle**.

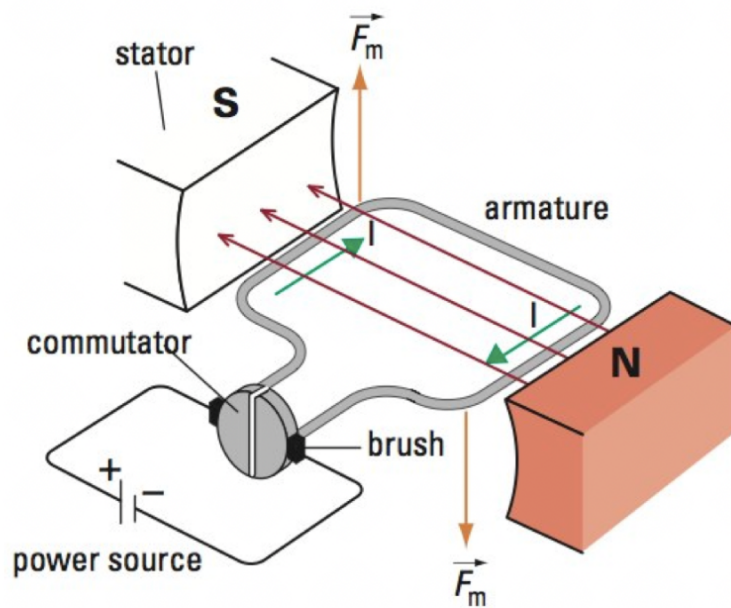
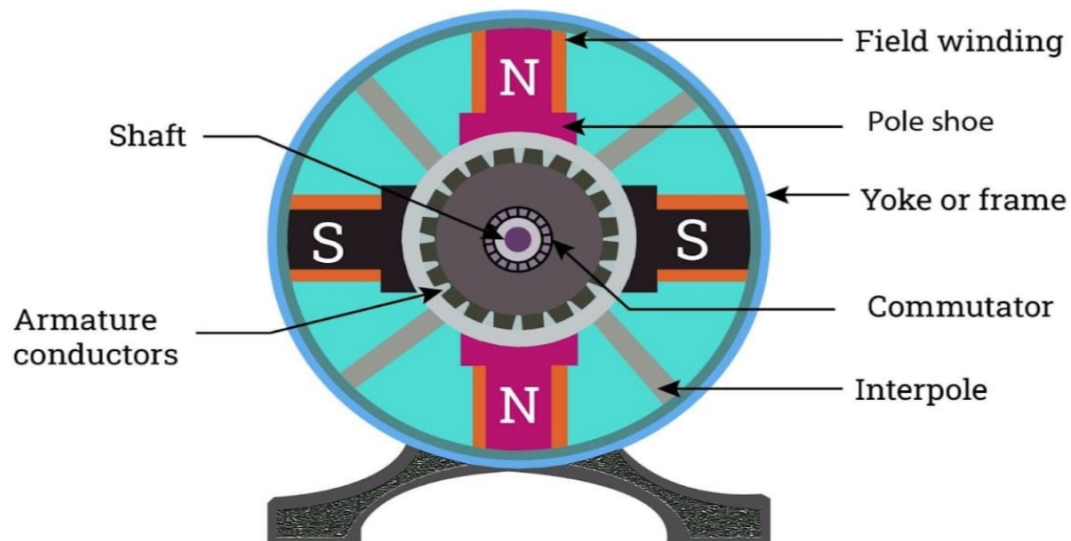
$$F = B i l \sin \theta$$

where:

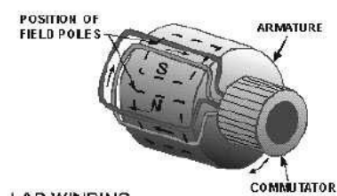
- (F) = force on conductor
- (B) = flux density (Wb/m²) **[MF]**
- (I) = current (A)
- (l) = length of conductor in field
- (θ) = angle between conductor and field
- The direction of force is given by **Fleming's Left-Hand Rule**.

When many conductors on the armature carry current, the combined effect produces a **torque** that causes rotation.

DC Motor Construction Parts



(A)



LAP WINDING

(B)



WAVE WINDING

3. Construction of D.C. Motor

1. Yoke [Basically Frame]

- Outer frame of the motor.
- Provides protection and mechanical support.
- Completes magnetic circuit.
- Made of cast steel or rolled steel.

2. Poles and Pole Shoes

- Poles carry field winding.
- Pole shoes spread flux uniformly and reduce reluctance.
- Made of laminated soft steel.

3. Field Windings

- Copper wire wound on poles.
- Provides magnetic field when excited with DC.
- Can be **shunt** (many turns, high resistance) or **series** (few turns, low resistance).

shunt $\rightarrow N \uparrow R \uparrow$
series $\rightarrow N \downarrow R \downarrow$

4. Armature Core

- Cylindrical, laminated to reduce eddy currents.
- Provides slots for armature conductors.

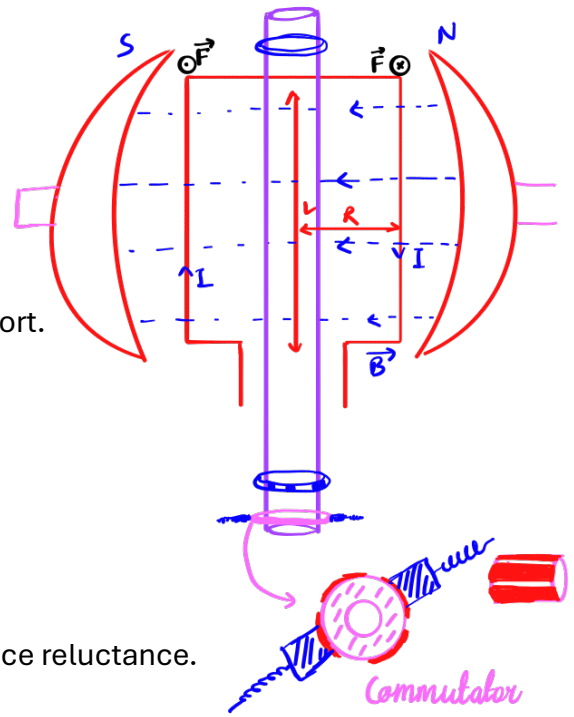
5. Armature Winding

- Copper conductors placed in slots.
- Types: lap winding (high current, low voltage), wave winding (low current, high voltage).

6. Commutator

- Cylindrical structure of copper segments, insulated with mica.
- Converts AC induced in armature into DC at terminals.
- Ensures unidirectional torque.

7. Brushes



- Made of carbon/graphite.
- Collect current from commutator and deliver it to external circuit.

4. Voltage Equation of a DC Motor

When motor runs:

- Back emf (V_B) is induced in the armature.
- It always opposes supply voltage (V).

$$V = V_B + I_a R_a$$

$$\Gamma = 2(R \times F)$$

$$\Gamma = 2BILR \sin \alpha$$

where:

- (V) = supply voltage
- (V_B) = back emf
- (I_a) = armature current
- (R_a) = armature resistance
- (Γ) = Torque

5. EMF Equation

The induced emf (same as generator emf) is:

$$V_B = \phi Z N \left(\frac{P}{A} \right)$$

where:

- (P) = number of poles
- (ϕ) = flux per pole (Wb)
- (Z) = total armature conductors
- (N) = speed of motor (rps)
- (A) = number of parallel paths

6. Power Equation

Input power to armature = $V I_a$

But using voltage equation:

$$V I_a = V_b I_a + I_a^2 R_a$$

- $V_b I_a$ = mechanical power developed in armature
- $I_a^2 R_a$ = copper loss in armature

So,

$$P = V_b I_a$$

7. Torque in DC Motor

- Power developed = torque $\times$ angular speed

$$P = \Gamma \cdot \omega$$

Where

$$\Gamma = \frac{1}{2\pi N} \cdot I_a \cdot \phi Z N (P/A)$$

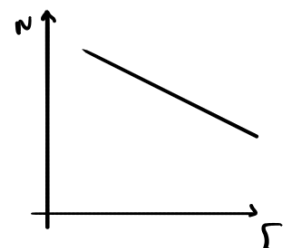
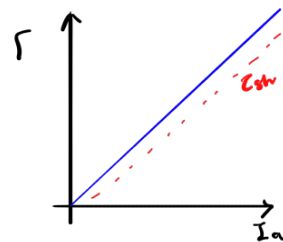
Therefore,

$$\Gamma = \frac{1}{2\pi} \phi Z (P/A) \cdot I_a$$

or more directly,

$$\Gamma \propto \phi I_a$$

Thus, torque is directly proportional to flux and armature current.



8. Back EMF (V_b) and Its Role

- Back emf is the **voltage induced** in the armature due to its rotation in the magnetic field.
- It **opposes the supply voltage** (Lenz's law).
- Functions as a **self-regulating mechanism**:
 - If load increases $\rightarrow$ motor slows $\rightarrow V_b$ decreases $\rightarrow$ more current flows $\rightarrow$ more torque is developed.
 - If load decreases $\rightarrow$ speed increases $\rightarrow V_b$ rises $\rightarrow$ current decreases $\rightarrow$ torque reduces.

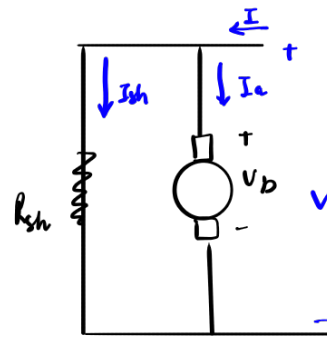
Hence, **motor adjusts automatically to load variations**.

9. Types of DC Motors

Based on field winding connections:

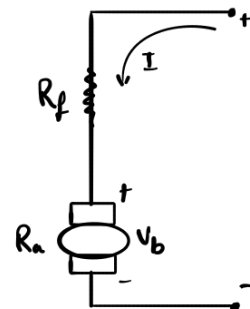
(i) Shunt Motor

- Field winding in parallel with armature.
- Characteristics:
 - Almost constant speed.
 - Torque proportional to armature current.
 - Moderate starting torque.
- Applications: fans, blowers, lathes, centrifugal pumps.



(ii) Series Motor

- Field winding in series with armature.
- Characteristics:
 - Very high starting torque $\tau \propto I_a^2$
 - Speed decreases with increase in load.
 - Must never run without load $\rightarrow$ can overspeed dangerously.

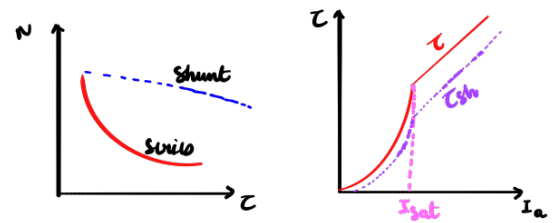


for series motor $\phi \propto I_a$

$$\tau \propto \phi \times I_a$$

$\tau \propto I_a^2$

- Applications: cranes, hoists, traction, elevators.



(iii) Compound Motor

- Has both series and shunt field windings.
- Two types:
 - **Cumulative Compound**: Series field aids shunt field → better starting torque + fairly constant speed.
 - **Differential Compound**: Series field opposes shunt field (rarely used, unstable).
- Applications: rolling mills, punching presses, heavy machinery.

10. Motor Characteristics

Three most important characteristics:

(i) Torque–Armature Current (T vs Ia)

- **Shunt motor**: straight line, $T \propto I_a$.
- **Series motor**: $T \propto I_a^2$ → very steep curve (high starting torque).
- **Compound motor**: lies between shunt & series.

(ii) Speed–Armature Current (N vs Ia)

- **Shunt motor**: nearly constant.
- **Series motor**: very high at no-load, decreases rapidly with load.
- **Compound motor**: moderate regulation.

(iii) Speed–Torque (N vs T)

- **Shunt motor**: flat characteristic.
- **Series motor**: sharp fall in speed with increase in torque.
- **Compound motor**: between the two.

11. Losses in DC Motor

1. Copper losses

- Armature copper loss = $I_a^2 R_a$
- Shunt field copper loss = $I_{sh}^2 R_{sh}$
- Series field copper loss = $I_{se}^2 R_{se}$

2. Iron losses

- Hysteresis loss = $k_h B^{1.6} f V$
- Eddy current loss = $k_e B^2 f^2 t^2 V$

3. Mechanical losses

- Bearing friction, brush friction, windage.

4. Stray load losses (approx 1% of output).

12. Efficiency of DC Motor

$$\eta = \frac{\text{output Power}}{\text{input Power}} \times 100$$

Max eff $\Rightarrow$ variable loss = Const losses

13. Applications of DC Motors

- **Shunt motors:** lathes, fans, centrifugal pumps, blowers, textile machinery.
 - **Series motors:** traction (railways), cranes, hoists, elevators, rolling mills.
 - **Compound motors:** punching machines, heavy presses, shears, rolling mills.
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14. Advantages of DC Motors

- High starting torque (especially series motors). **[At low RPM]**
- Speed control is easy and wide.
- Reliable and robust.

15. Limitations

- Require DC supply or rectifiers (in AC systems).
 - Brushes and commutator need regular maintenance. **[Friction]**
 - Less efficient compared to modern AC induction motors.
 - **use of Brars**
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